Fleet condition report: Kelmarsh wind farm

Kelmarsh wind farm (6x Senvion MM92, UK), Cubico, Greenbyte export. 6 turbines, SCADA 21 Jan 2016 to 31 Dec 2024; assessment window: the last 24 months of each turbine (Jan 2023 to Dec 2024). Generated by `windml report` from the mapping `kelmarsh_all_years.yaml`.

Executive summary

All 6 turbines are green: no validated warning sign in the data up to 31 Dec 2024. Green is not a guarantee; see the notes and limits below.

Green: KEL01, KEL02, KEL03, KEL04, KEL05, KEL06.

Worth knowing. Shown for context only: these signals (model alerts, past availability) did not reliably precede events in the validation, so they do not change the rating.

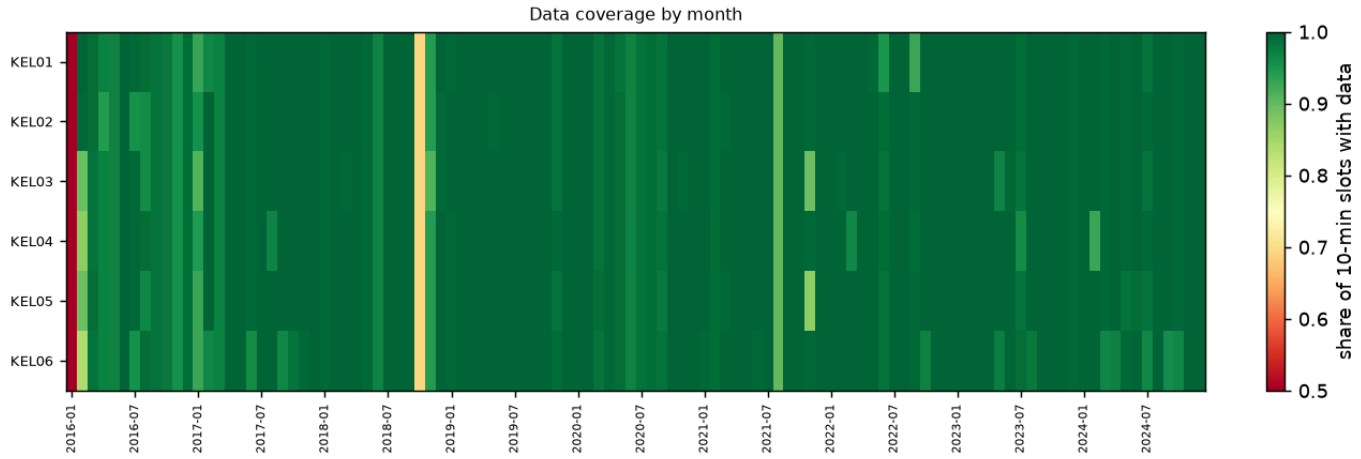
- **KEL01**: Its pitch system behaved unlike its own normal-behaviour model on 36% of days in the last 3 months.
- **KEL05**: Its pitch system behaved unlike its own normal-behaviour model on 47% of days in the last 3 months.
- **KEL06**: Availability was 89.6% over the last 6 months vs 99.3% for the other turbines; largest fault cause: generator (234 h). This is past downtime, not a forecast.

How the ratings work. A turbine is amber only for a specific, unusual pattern: gear-oil pump pressure building up far beyond its sister turbines, repeated oil-system or pitch trips, or a faulty sensor. It is red when the pattern is strong (e.g. a pump pressure rise of 2 bar or more) or persistent and still active. Thresholds were chosen on 2017-2020 data from two UK wind farms and checked on 2021-2024: amber or red turbine-months were followed by a logged fault, trip or long service stop within 90 days 76% of the time, against 50% for random turbines in the same months; but only 19% of the turbine-months before such an event carried a flag. Ratings help decide where to look first; most faults still arrive without a warning sign in SCADA data.

How sure are we. Confidence is *High* when the turbine has 3+ years of history, near-complete data, models trained on earlier years, an alarm log and at least 3 sister turbines to compare with; it drops when any of these is missing. A red rating is a reason to look, not a diagnosis.

1. Data quality

Turbine	First data	Last data	Months	Coverage (window)	Longest gap (h)	Signals	Native interval	Frozen rows (window)	Stuck sensors (>= 1%)	Stop/fault labels
KEL01	2016-01-23	2024-12-31	107	99.8%	6	9	10min	0	-	alarm log
KEL02	2016-01-21	2024-12-31	107	99.8%	6	9	10min	0	-	alarm log
KEL03	2016-01-27	2024-12-31	107	99.6%	25	9	10min	0	-	alarm log
KEL04	2016-02-04	2024-12-31	107	99.4%	27	9	10min	3,196	gear_oil_inlet_temp_c, gear_oil_temp_c	alarm log
KEL05	2016-01-24	2024-12-31	107	99.7%	6	9	10min	0	-	alarm log
KEL06	2016-02-05	2024-12-31	107	98.9%	28	9	10min	0	-	alarm log



Signals present in at least one turbine: `ambient_temp_c`, `available_capacity_kw`, `converter_inlet_temp_c`, `data_availability`, `energy_export_kwh`, `gear_oil_inlet_pressure_bar`, `gear_oil_inlet_temp_c`, `gear_oil_pump_pressure_bar`, `gear_oil_temp_c`, `gen_bearing_front_temp_c`, `gen_bearing_rear_temp_c`, `generator_speed_rpm`, `hub_temp_c`, `lost_prod_curtail_total_kwh`, `lost_prod_downtime_kwh`, `main_bearing_front_temp_c`, `main_bearing_rear_temp_c`, `nacelle_position_deg`, `nacelle_temp_c`, `pitch_a_deg`, `pitch_b_deg`, `pitch_c_deg`, `pitch_motor_temp_1_c`, `pitch_motor_temp_2_c`, `pitch_motor_temp_3_c`, `power_kw`, `power_setpoint_kw`, `rotor_bearing_temp_c`, `rotor_speed_rpm`, `wind_direction_deg`, `wind_speed_1_ms`, `wind_speed_2_ms`, `wind_speed_ms`, `yaw_bearing_angle_deg`.

2. Turbine scorecard

Turbine	Flag	Confidence	Tech. avail.	Energy avail.	Fault h/yr	Largest fault cause	Oil ep.	Pitch ep.	Abnormal oil days, final yr	Under-perf. days, final yr	Pump pressure: base to last (bar)	Oil-temp model error
KEL01	GREEN	High	99.5%	99.2%	4	hydraulics (4 h)	0	1	8%	0%	7.2 to 7.0	1.7x
KEL02	GREEN	High	99.4%	98.9%	9	pitch (10 h)	0	3	2%	0%	6.6 to 7.1	1.3x
KEL03	GREEN	High	98.6%	98.5%	71	control_comm (96 h)	0	1	2%	0%	7.0 to 7.2	1.4x
KEL04	GREEN	High	93.8%	92.1%	330	converter (407 h)	2	4	15%	0%	7.0 to 7.4	3.1x
KEL05	GREEN	High	95.2%	96.2%	354	gearbox (370 h)	1	11	5%	0%	6.9 to 7.4	1.7x
KEL06	GREEN	High	95.8%	95.7%	234	generator (234 h)	0	2	6%	0%	7.0 to 6.9	1.8x

Ratings, reasons and notes

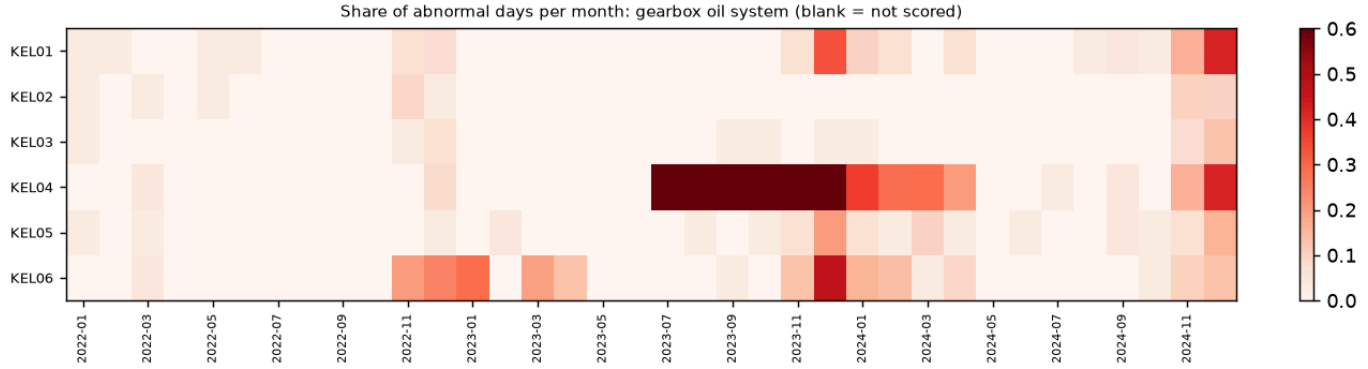
- **KEL01** GREEN (confidence High: 107 months of history, models trained on earlier years). No rating trigger. Notes: Its pitch system behaved unlike its own normal-behaviour model on 36% of days in the last 3 months.
- **KEL02** GREEN (confidence High: 107 months of history, models trained on earlier years). No rating trigger.
- **KEL03** GREEN (confidence High: 107 months of history, models trained on earlier years). No rating trigger.
- **KEL04** GREEN (confidence High: 107 months of history, models trained on earlier years). No rating trigger. Notes: Sensor(s) gear_oil_inlet_temp_c, gear_oil_temp_c were frozen for a while earlier in the window but read normally now.
- **KEL05** GREEN (confidence High: 107 months of history, models trained on earlier years). No rating trigger. Notes: Its pitch system behaved unlike its own normal-behaviour model on 47% of days in the last 3 months.
- **KEL06** GREEN (confidence High: 107 months of history, models trained on earlier years). No rating trigger. Notes: Availability was 89.6% over the last 6 months vs 99.3% for the other turbines; largest fault cause: generator (234 h). This is past downtime, not a forecast.

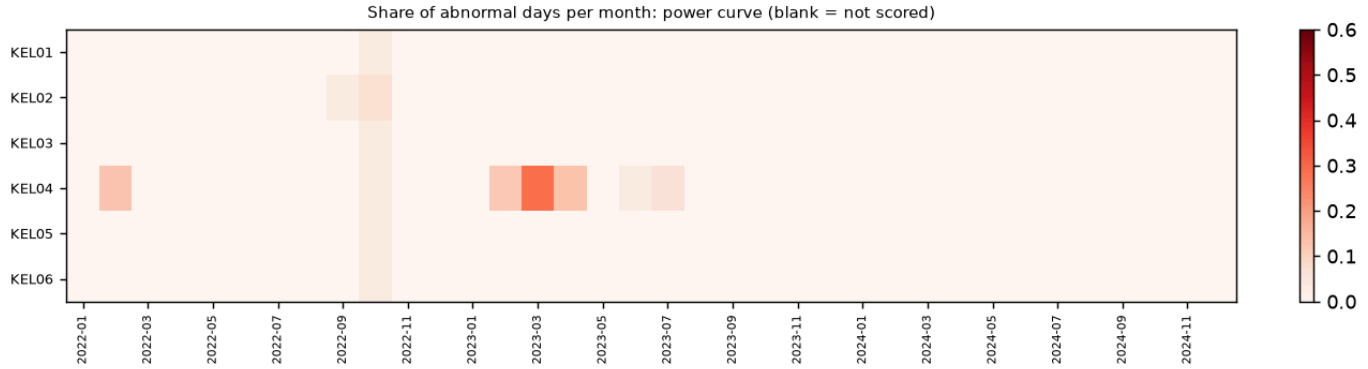
3. Model-based health (normal-behaviour residuals)

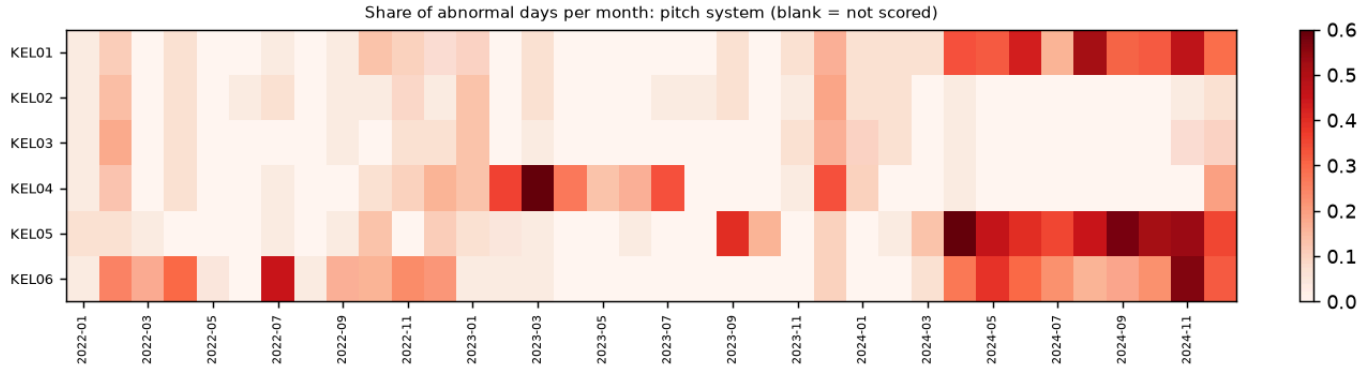
Each turbine gets its own models that predict a signal (oil temperature, bearing temperature, pitch angle, power...) from operating conditions. A day is *abnormal* when the turbine's median deviation from its own model is beyond 3 robust standard deviations in the harmful direction. *Chronic* means at least 30% abnormal days in a month.

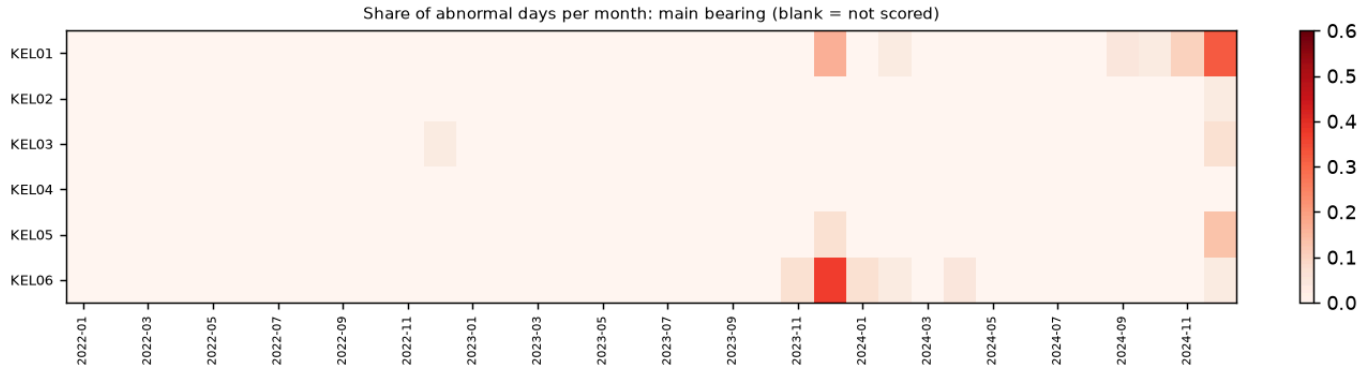
Turbine	Training scheme	Models	Window scored	gearbox oil system: abnormal days final yr / chronic months	main bearing: abnormal days final yr / chronic months	pitch system: abnormal days final yr / chronic months	power curve: abnormal days final yr / chronic months
KEL01	walk-forward	14	99%	8% / 2	4% / 1	28% / 7	0% / 0
KEL02	walk-forward	14	99%	2% / 0	0% / 0	2% / 0	0% / 0
KEL03	walk-forward	14	99%	2% / 0	1% / 0	3% / 0	0% / 0
KEL04	walk-forward	14	94%	15% / 8	0% / 0	3% / 4	0% / 0
KEL05	walk-forward	14	96%	5% / 0	1% / 0	37% / 10	0% / 0
KEL06	walk-forward	14	96%	6% / 1	1% / 1	23% / 4	0% / 0

Chronic abnormal turbines (3+ consecutive months with >= 30% abnormal days): KEL01 pitch system (4 months); KEL04 gearbox oil system (7 months); KEL05 pitch system (9 months).


Share of abnormal days per month: gearbox oil system (blank = not scored)
KEL01
KEL02
KEL03
KEL04
KEL05
KEL06
2022-01
2022-03
2022-05
2022-07
2022-09
2022-11
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
0.6
0.5
0.4
0.3
0.2
0.1
0.0

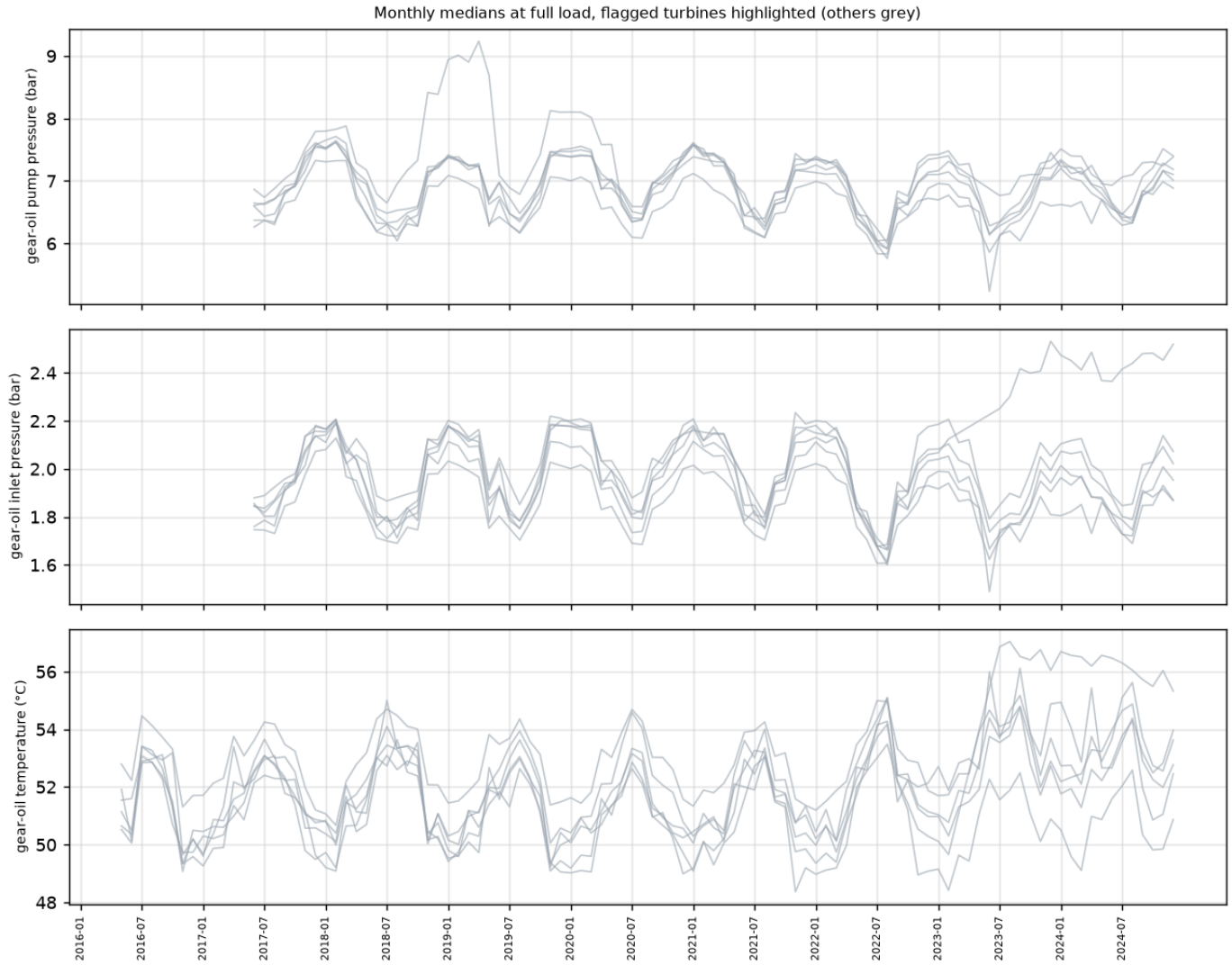

Share of abnormal days per month: power curve (blank = not scored)
KEL01
KEL02
KEL03
KEL04
KEL05
KEL06
2022-01
2022-03
2022-05
2022-07
2022-09
2022-11
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
0.6
0.5
0.4
0.3
0.2
0.1
0.0


Share of abnormal days per month: pitch system (blank = not scored)
KEL01
KEL02
KEL03
KEL04
KEL05
KEL06
2022-01
2022-03
2022-05
2022-07
2022-09
2022-11
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
0.6
0.5
0.4
0.3
0.2
0.1
0.0


Share of abnormal days per month: main bearing (blank = not scored)
KEL01
KEL02
KEL03
KEL04
KEL05
KEL06
2022-01
2022-03
2022-05
2022-07
2022-09
2022-11
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
0.6
0.5
0.4
0.3
0.2
0.1
0.0

4. Gearbox oil pressure and temperature trends

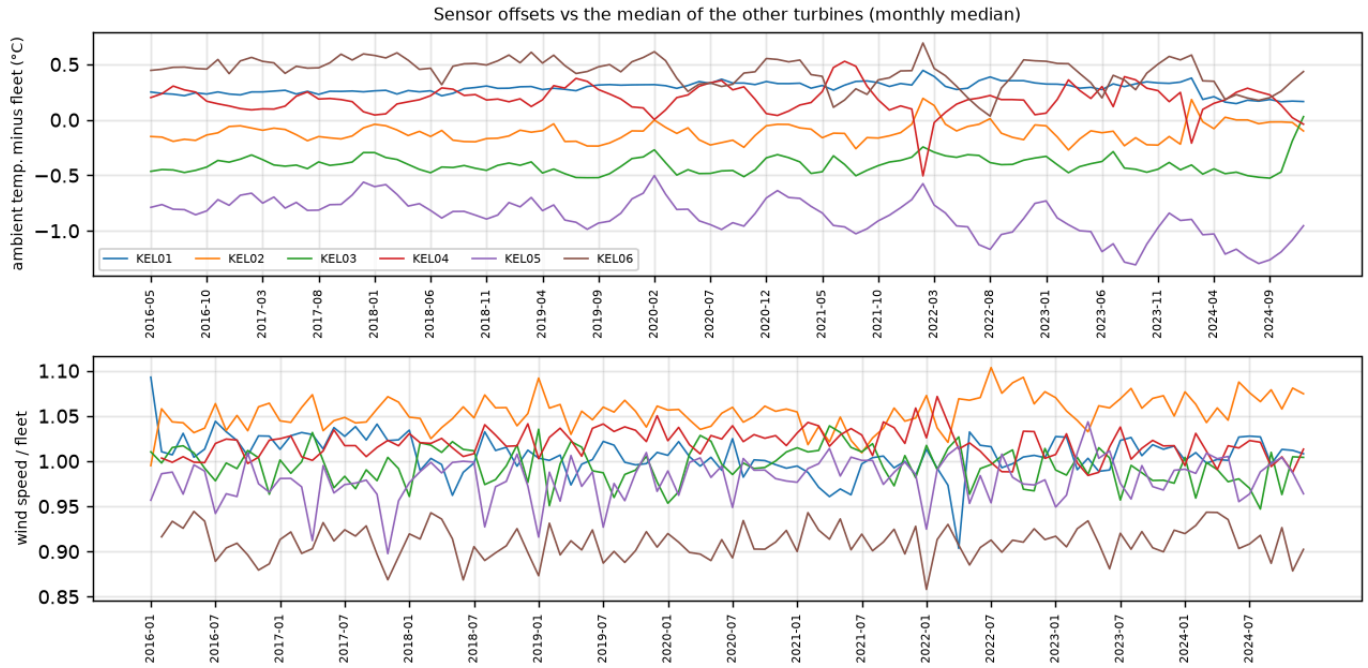
Turbine	Pump pressure (bar): baseline → last month	Inlet pressure (bar)	Oil temperature (°C)	Baseline months
KEL01	7.2 → 7.0 (-0.2)	1.9 → 1.9 (-0.1)	51.8 → 53.6 (+1.8)	2017-06..2022-12
KEL02	6.6 → 7.1 (+0.5)	1.9 → 2.0 (+0.1)	53.0 → 52.8 (-0.2)	2017-06..2022-12
KEL03	7.0 → 7.2 (+0.2)	2.0 → 2.0 (-0.0)	51.2 → 52.5 (+1.3)	2017-06..2022-12
KEL04	7.0 → 7.4 (+0.4)	2.0 → 2.5 (+0.5)	51.1 → 55.3 (+4.2)	2017-06..2022-12
KEL05	6.9 → 7.4 (+0.5)	2.0 → 2.1 (+0.1)	51.3 → 50.9 (-0.4)	2017-06..2022-12
KEL06	7.0 → 6.9 (-0.1)	2.0 → 1.9 (-0.2)	51.2 → 54.0 (+2.7)	2017-06..2022-12



5. Sensor drift and stuck sensors

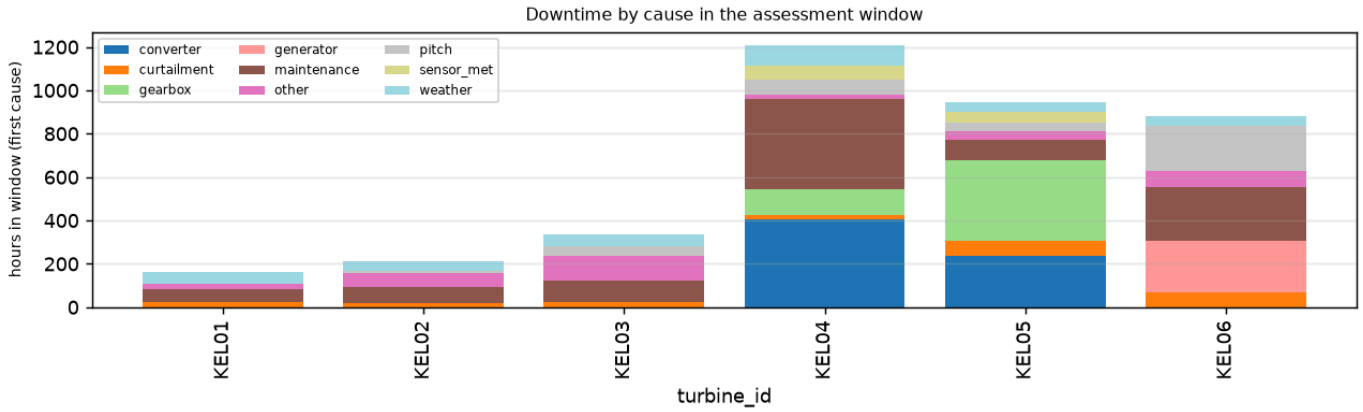
Drift = change of a turbine's offset to the median of the other turbines, last 3 months vs before the window (thresholds: 1.5 °C, 5% wind speed).

Turbine	drift ambient_temp_c	drift wind_speed_ms	Stuck sensors
KEL01	-0.12 °C	0.7%	-
KEL02	+0.10 °C	2.3%	-
KEL03	+0.22 °C	0.6%	-
KEL04	-0.16 °C	-1.8%	gear_oil_inlet_temp_c, gear_oil_temp_c
KEL05	-0.28 °C	0.6%	-
KEL06	-0.12 °C	-0.8%	-



6. Stop and fault history

Turbine	Fault stops	Maintenance	Grid	Weather	Curtailment	Major-fault episodes	Maint. h/yr	Oil / pitch trips
KEL01	25	36	15	4	4	1	31	0 / 6
KEL02	34	36	28	2	4	3	35	0 / 3
KEL03	7	56	14	3	4	1	50	0 / 1
KEL04	43	50	27	4	4	4	207	5 / 4
KEL05	236	47	14	2	4	10	47	1 / 47
KEL06	36	54	19	1	4	2	125	0 / 7



Most frequent fault stops in the window (all turbines):

Code	Message	Component	Stops
6530	Anemometer defect	sensor_met	175
455	Repeating error BP52	pitch	56
3110	Frequency converter error	converter	31
4540	Tower oscillation X level 2	tower	26
4510	Tower oscillation Y level 1	tower	20
4520	Tower oscillation X level 1	tower	18
1070	Drive train monitor level 2	gearbox	6
1620	Implausible gear speed	gearbox	5
2500	Maximum generator speed	generator	4
4530	Tower oscillation Y level 2	tower	4
570	Pitch too slow BP180	pitch	3
1510	Low gearbox oil pressure	gearbox	3

7. Methods, thresholds and limits

- **Pipeline.** `windml report` reads the raw export only through the mapping (`windml.ingest`), cleans and labels each turbine (QC flags, short-gap interpolation, alarm-log stop labels), trains the models, computes the metrics and writes this report. Every number here is reproducible from the mapping and the raw files.
- **Window.** The last 24 months of each turbine; *final year* = its last 12 months.
- **Models.** LightGBM per turbine and signal on healthy operating rows (no stop within 6 h, no fault, not in the 30 days before a major fault, QC clean). With 2+ calendar years, each year is scored by a model trained only on earlier years (3-day embargo, training must cover all 12 months). With 12-23 months, the record is cut into 6 blocks scored by models trained on the other blocks (cross-fit). Under 12 months, no model.
- **Ratings.** Computed at each month-end from data up to that date (latest month-end shown here). AMBER when: the full-load gear-oil pump pressure (median of the last 15 days) is >= 1 bar above the turbine's own baseline (days 60-365 before) AND >= 5 robust standard deviations above the other turbines' rise that month; >= 3 oil-system trip episodes in 6 months; >= 6 pitch trip episodes in 6 months; an ambient-temperature or wind-speed sensor drifting vs the fleet over the last 3 months (1.5 °C, 5%); or a monitored sensor frozen on more than 2% of the last 90 days. RED when the pump rise is >= 2 bar; or a trip rule has held 2 month-ends in a row with >= 3x the amber count and a trip in the last 30 days; or two independent systems are amber at once. Shown as notes but never colour-setting, because they had no reliable lead on events in the validation: low availability (it describes the past), model-based abnormal-day rates (oil, pitch, main bearing, power curve) and missing data. A trip episode merges trips less than 24 h apart.
- **Validation.** Retrospective check on Kelmarsh and Penmanshiel (21 turbines). Thresholds selected on 2017-2020, checked on 2021-2024 (792 turbine-months): pump build-up → oil trip or service stop: 58% of 12 flagged turbine-months vs 55% by chance; repeat oil trips → further oil trips: 63% of 75 flagged turbine-months vs 32% by chance; repeat pitch trips → further pitch trips: 67% of 6 flagged turbine-months vs 36% by chance. Any amber or red: 76% followed by a logged event within 90 days vs 50% by chance (recall 19%); red alone 65% vs 58% on only 17 turbine-months, so red marks strength and persistence more than a higher probability. Held-out ratings: 89% green, 9% amber, 2% red. Consecutive months overlap, so the effective sample is much smaller than the row counts; details in `reports/fleet_proof/calibration.md`.
- **Confidence.** Points for history (3+ years: 2, 2-3 years: 1), coverage >= 95%, models trained on earlier years, an alarm log, and a fleet of 4+ turbines (peer comparison). High >= 5, Medium 3-4, Low <= 2.
- **Limits.** The alarm-log classes come from the mapping's rules; trip detection uses Senvion-style oil/pitch codes and messages (other vendors need `class_rules`). Full-load oil medians assume one oil circuit design across the fleet. Models learn whatever was normal in their training data, including a fault that was already present. Sensor drift is relative to the other turbines, so a fleet-wide drift is invisible. The rating thresholds were calibrated on two UK farms of one turbine family (Senvion MM92/MM82); on other fleets the peer-relative logic still applies but the hit rates are unknown. Ratings look back up to 6 months and only at what SCADA and the alarm log show: inspections, oil analyses, vibration (CMS) data and work orders are not used, so a recent repair or a fault the sensors cannot see changes nothing here.
- **Data.** CC BY 4.0, Cubico Sustainable Investments, doi:10.5281/zenodo.16807551.